

Detecting cholera, typhoid & polio in wastewater — up to 2 weeks before hospitals are overwhelmed

THE PROBLEM

In non-sewered West African settings, hospital surveillance detects outbreaks **2–4 weeks after** community spread begins. By then, response is reactive and costly.

OUR SOLUTION

- Metagenomic sequencing of wastewater from canals, drains & markets — **no sewer network required**
- Multi-pathogen detection in a single sample: *V. cholerae*, *S. typhi*, Poliovirus, AMR
- PMMoV/CrAssphage normalization for cross-site comparability
- Open-source Snakemake pipeline + real-time alert dashboard
- PHA4GE-compliant data for continental interoperability

WHY IT MATTERS FOR AFRICA CDC

- First fully **open-source, sovereign** WBE pipeline for non-sewered West Africa
- Aligned with Africa CDC Continental WES Strategy (GLOWACON)
- Designed for replication — any country with a sequencer can deploy it

VS. EXISTING PROGRAMS

Feature	Burkina	AquaVigil
In-country sequencing	✗	✓
Open pipeline (GitHub)	✗	✓
Multi-pathogen / sample	✗	✓
Non-sewered environments	✗	✓
Weekly real-time alerts	✗	✓

PILOT & TIMELINE

5 sampling sites across Lomé, Togo. Simulated-data validation (Aug 2026) → real African datasets from NCBI SRA (Sep 2026) → field samples from Lomé (Oct 2026) → bioRxiv preprint (Oct 2026) → **CPHIA presentation, Addis-Ababa (Nov 2026).**

Our Ask: (1) A 30-min introductory call with your WES/WBE focal point · (2) Guidance on alignment with the Africa CDC WES framework · (3) Consideration for a CPHIA 2026 presentation slot.